CS & IT ENGINEERING



Algorithms

Dynamic Programming



Lecture No.- 07

By- Aditya sir

Recap of Previous Lecture



- Topic
- Topic

MCM

SOS → similar to 0/1 Knapsack
↳ Decision Problem

Topics to be Covered



- Topic
- Topic
- Topic

Multi-stage graph

TSP



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on Linkedin where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

Multi-Stage Graph :



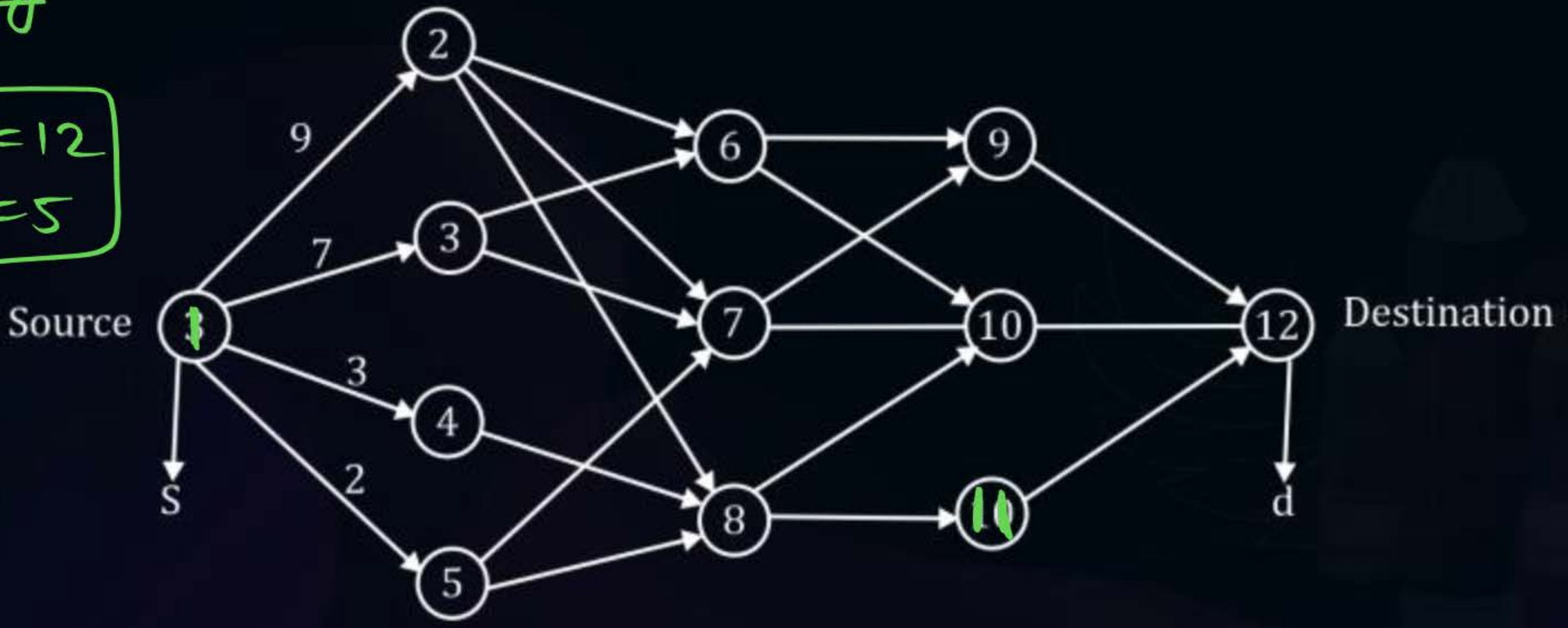
Topic : Dynamic Programming :(DP)

Eg.1.

$V_i \longrightarrow V_{i+1}$

Stage $\longrightarrow$ V_1 V_2 V_3 V_4 V_5

$n = 12$
$l = 5$





Topic : Dynamic Programming :(DP)

Multi- Stage graph:

Given: G(V,E)

$|V| = n$

$|E| = e$

There are l- stages

The vertex in any stage i can only be connected to stage i +1.

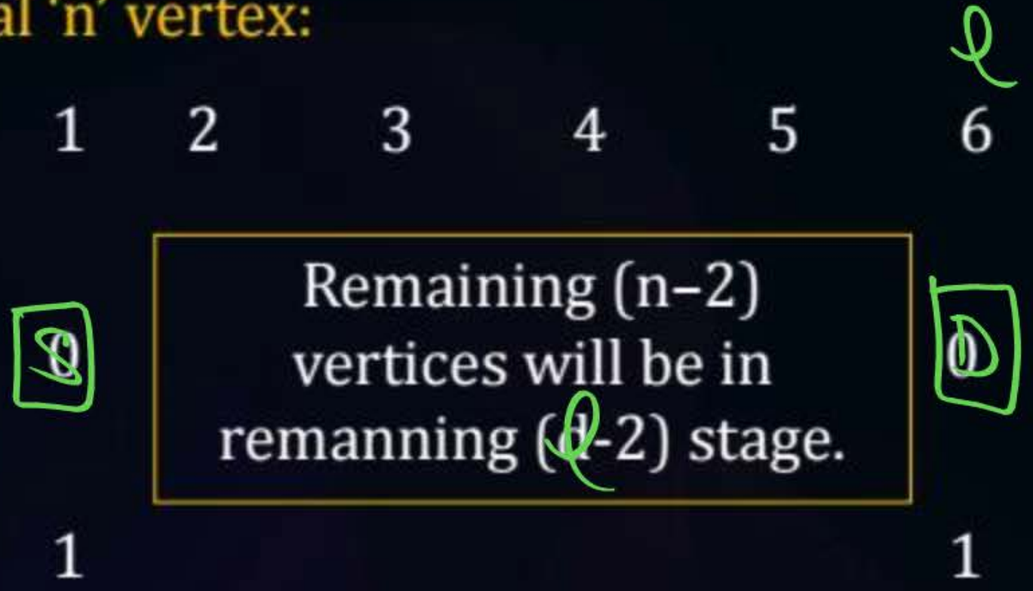
$V_i \rightarrow V_{i+1}$



Topic : Dynamic Programming :(DP)

Stage -1 and stage-4 (last stage) both will have only 1 vertex each.

Total 'n' vertex:





Topic : Dynamic Programming :(DP)

	1 2 3 4 5.....12
1	
2	
3	→ Cost adjacency Matrix
4	
5	C [i , j] → edge cost
.	e.g. i→ j
.	
.	
.	
12	



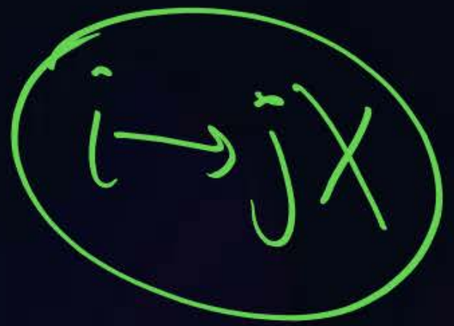
Topic : Dynamic Programming :(DP)

Logic: (V. V. V. imp)

Let cost (i, j) represent cost of the path form vertex 'j' present in stage 'i' to each the destination vertex 'd'.

↓ ↓

Cost (i , j)





Topic : Dynamic Programming :(DP)

Overall problem: Cost (1,1)

Cost of the path from a vertex 1 present in stage 1 to reach the destination vertex 'd'.

Cost (i, j)

- vertex 'j' present in stage 'i'
- Stage number

Topic : Dynamic Programming :(DP)

Source: 1

$v_i \rightarrow$ Stage

Cost (1,1) = min

$K \in V_2$

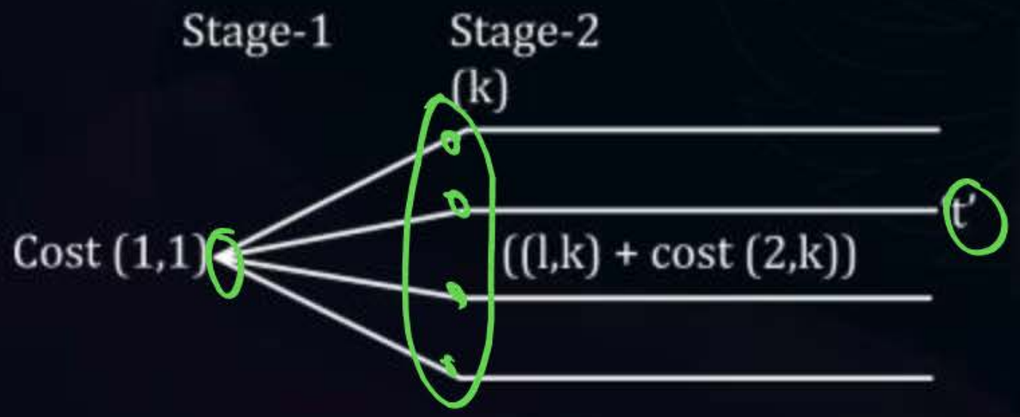
Stage-2 and

$< 1, k > \in E$

edge cost $1 \rightarrow K$

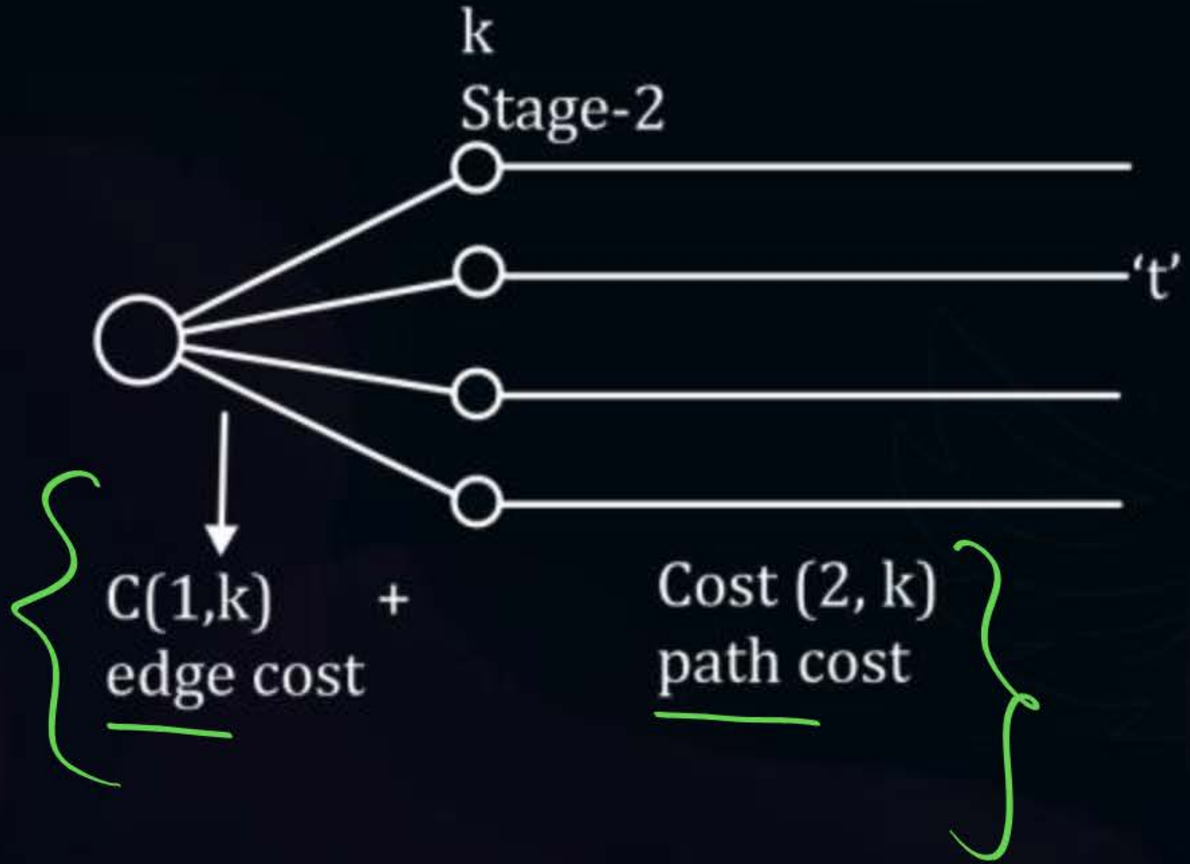
$C(1, k) + cost(2, k)$

Subproblem



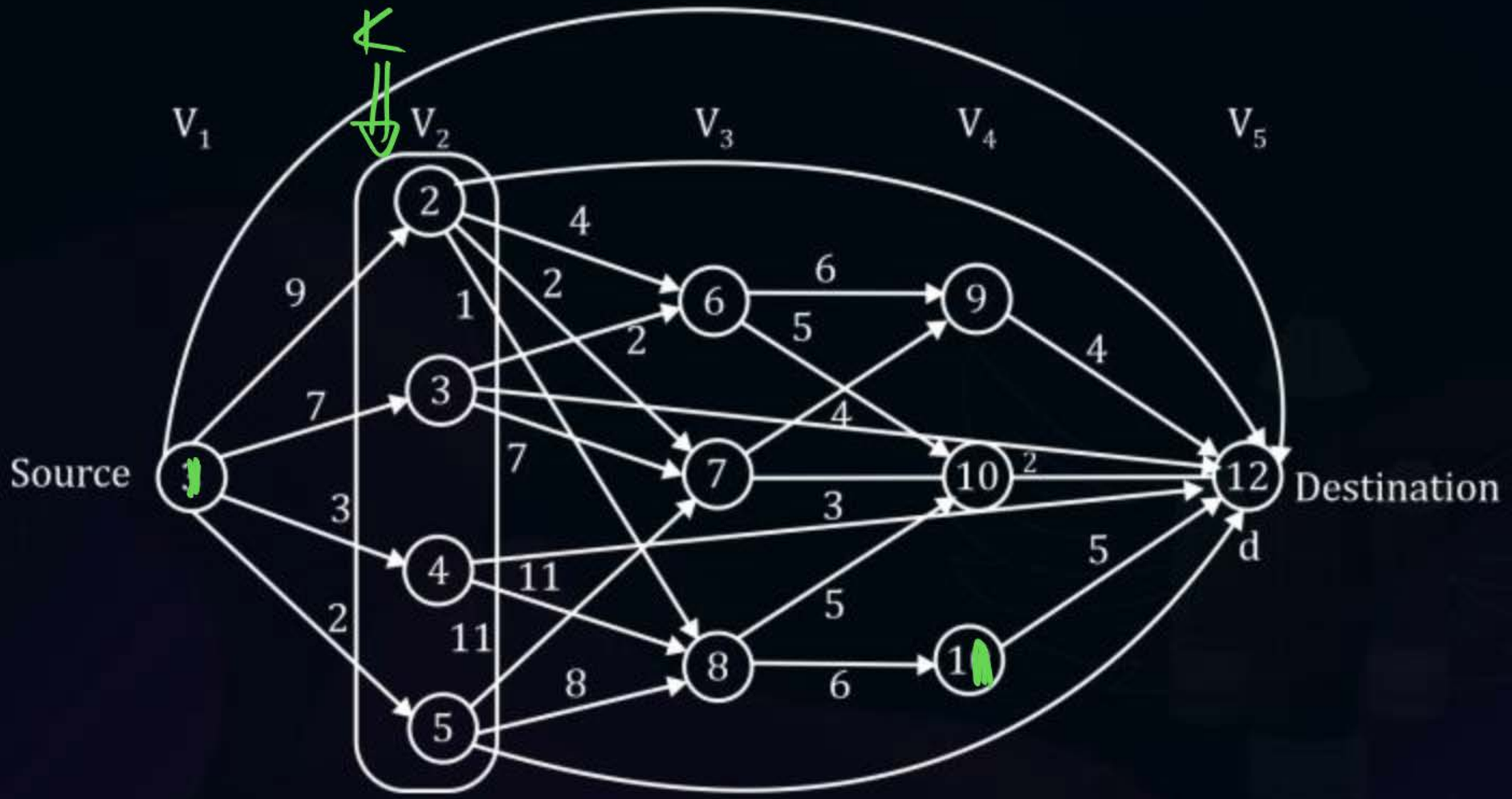


Topic : Dynamic Programming :(DP)



Topic : Dynamic Programming :(DP)

Eg. 4.





Topic : Dynamic Programming :(DP)

Optimization Problem

Objective function

What criteria is to minimized/ maximized

(SPSP)

Minimize the cost of the path from source → Destination



Topic : Dynamic Programming :(DP)

1. C(i,j): cost of edge from i^{th} vertex to j^{th} vertex

   $i \rightarrow j$

2. Cost (i,j) = cost of path from vertex 'j' present in stage 'i' to reach the destination 't'.

IAm



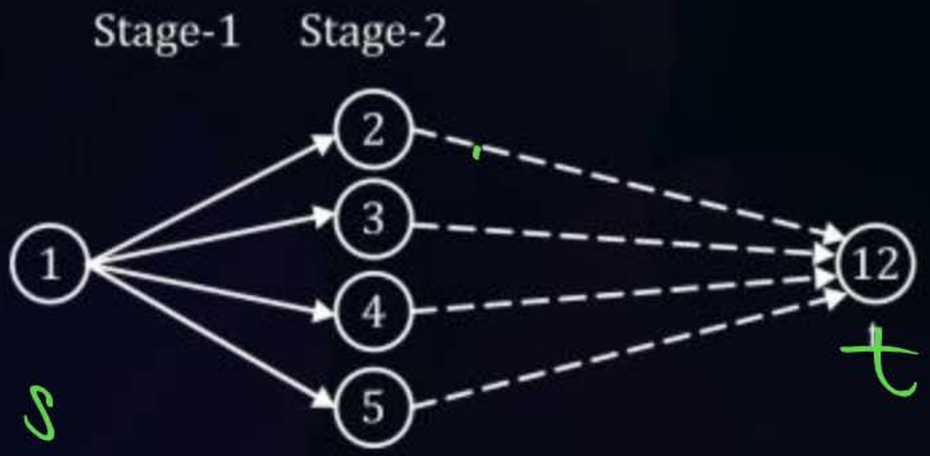
Topic : Dynamic Programming :(DP)

Multi-stage graph:

Source → Destination

(s) (t)

Shortest path between 's' to 't'



S1 S2 Sl

1 → k ⇝ t

edge + Shortest path (k → t)

1 → k Subproblem

$K \in V_2$ Cost (2, k)

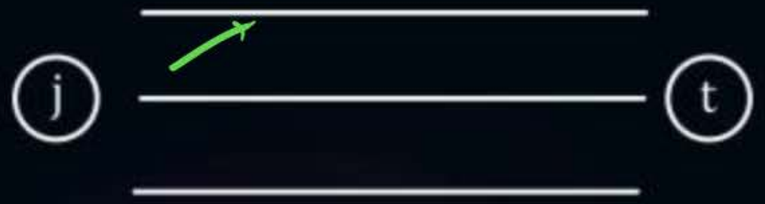
and

(1, k) ∈ E



Topic : Dynamic Programming :(DP)

Generalized Recurrence for multi-stage graph problem. 'i'



$$cost(i, j) = \min \{c(j, k) + cost(i + 1, k)\}$$

Stage number

Vertex and $(j, k) \in E$



Base-Condition:

2nd last stage $(l-1)$

$cost(l-1, j) = c(j, t)$



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Recurrence:

Cost (i, j) = Min {c (j, k) + cost (i +1, k)}....(1)

$K \in V_{i+1}$ and (j, k) $\in$ E

l = total stager

Cost (l -1, j) = c (j, t).....(2)

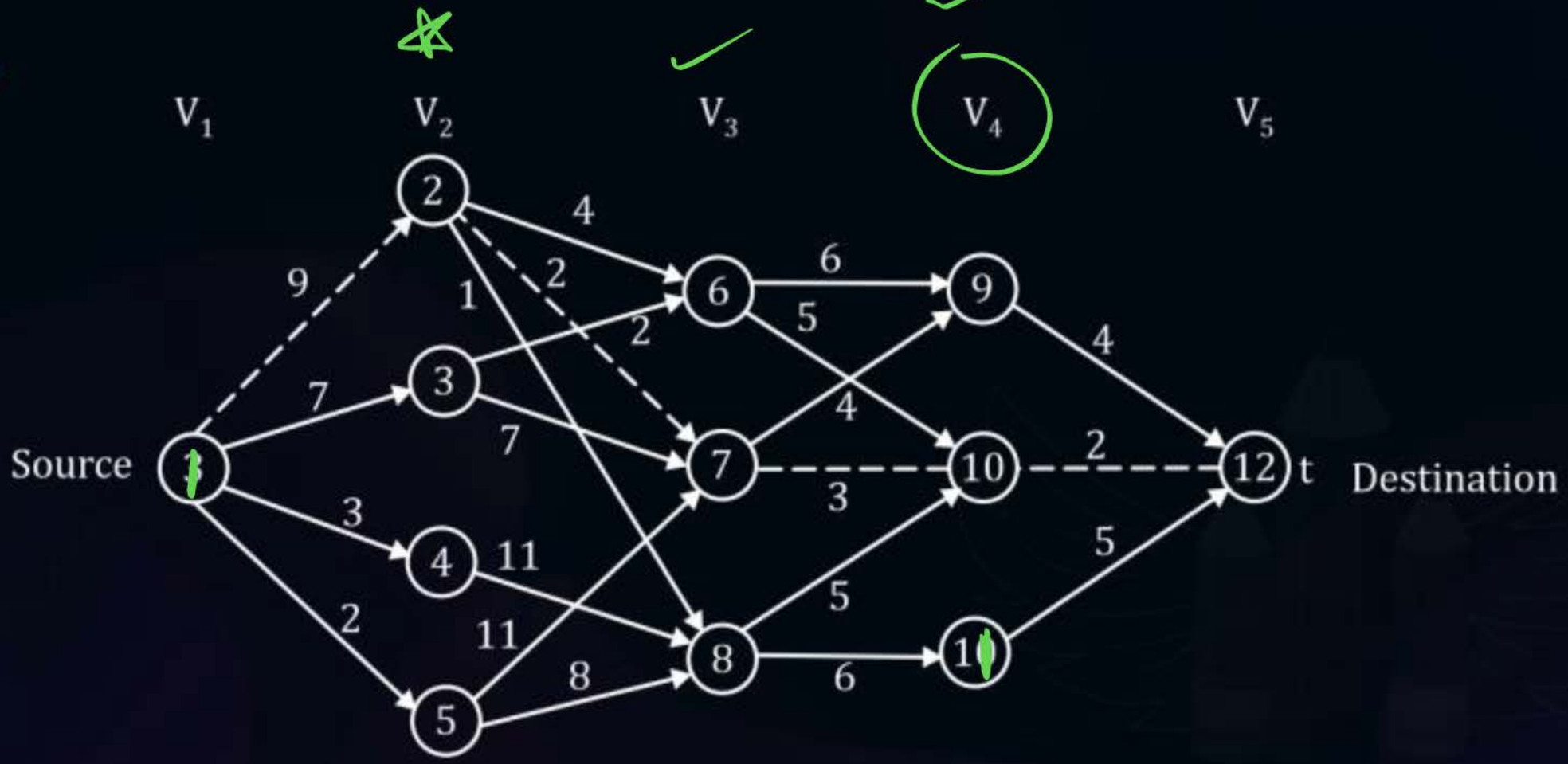
D(i,j) = 'k' that minimizes equation.



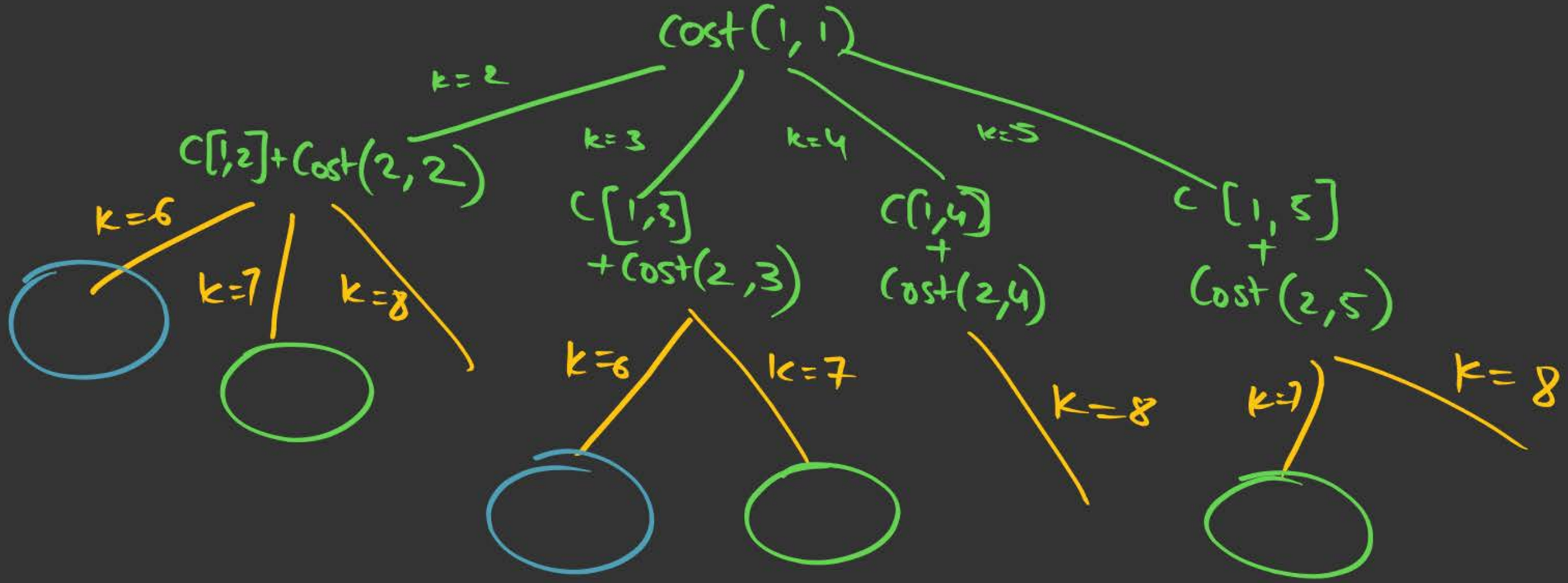


Topic : Dynamic Programming :(DP)

E.g. 1.



Top-Down

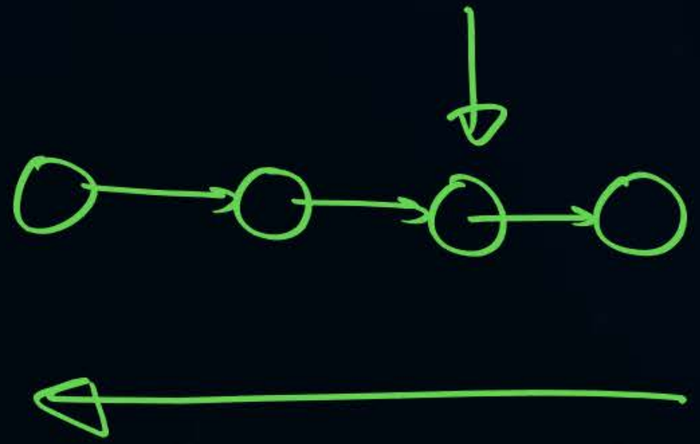


overlapping subproblem



Topic : Dynamic Programming :(DP)

Stage-1



Bottom UP

$$\text{Cost}(1,1) = \min \begin{cases} K = 2 : c(1,2) + \text{cost}(2,2) \\ K = 3 : c(1,3) + \text{cost}(2,3) \\ K = 4 : c(1,4) + \text{cost}(2,4) \\ K = 5 : c(1,5) + \text{cost}(2,5) \end{cases}$$



Topic : Dynamic Programming :(DP)

Stage-4

(2nd last stage)

K = 9, 10, 11

D (4, 9) = t

Cost (4, 9) = c (9, t) =4

D (4, 10) = t

Cost (4, 10) = c (10, t) =2

D (4, 11) = t

Cost (4, 11) = c (11, t) =5

Stage-1
↓
Stage-2
↓
Stage-3
↓
Stage-4



Topic : Dynamic Programming :(DP)

Stage-3:

J = 6,7,8

Cost (3, j)

$$\text{Cost}(3,6) = \text{Min}\begin{cases} K = 9: & C(6,9) + \text{Cost}(4,9) \\ k = 10: & C(6,10) + \text{Cost}(4,10) \end{cases}$$

$$= \text{Min}\begin{cases} K = 9: & 6 + 4 \\ K = 10: & 5 + 2 \end{cases}$$ Pre. Stage result

D(3, 6) = 10

$$= \text{Min}\begin{cases} K = 9: & 10 \\ K = 10: & 7 \end{cases} \Rightarrow 7$$



Topic : Dynamic Programming :(DP)

$$\text{Cost}(3,7) = \begin{cases} K = 9: & C(7,9) + \text{Cost}(4,9) \\ K = 10: & C(7,10) + \text{Cost}(4,10) \end{cases}$$

$$\begin{cases} K = 9: & 4 + 4 \\ K = 10; & 3 + 2 \end{cases}$$

D(3,7)=10

$$= \text{Min} \begin{cases} K = 9: & 8 \\ K = 10; & 5 \end{cases} \Rightarrow 5$$



Topic : Dynamic Programming :(DP)

Cost (3,8)= Min { K = 10: C (8, 10) + Cost (10, 10)
K = 11: c (8, 11) + Cost (4, 11) }

Min { K = 10: 5 + 2
K = 11: 6 + 5 }

Min { K = 10: 7
K = 11: 11 } = 7

D (3, 8) = 10



Topic : Dynamic Programming :(DP)

Stage-2: J = 2, 3, 4, 5

$$\text{cost}(2,2) = \text{Min}\begin{cases} K=6: C(2,6) + \text{Cos}(3,6) \\ K=7: C(2,7) + \text{Cos}(3,7) \\ K=8: C(2,8) + \text{Cos}(3,8) \end{cases}$$

$$= \text{Min}\begin{cases} K=6: 4+7 \\ K=7: 2+5 \\ K=8: 1+7 \end{cases}$$

$$D(2,2) = 7 \qquad = \text{Min}\begin{cases} K=6: 11 \\ K=7: 7 \\ K=8: 8 \end{cases} = 7$$



Topic : Dynamic Programming :(DP)

$$\text{Cost}(2,3) = \min \begin{cases} K = 6 : C(3,6) + \text{Cost}(3,6) \\ K = 7 : C(3,7) + \text{cost}(3,7) \end{cases}$$

$$D(2,3) = 6 \quad \min \begin{cases} K = 6 : 2 + 7 \\ K = 7 : 7 + 5 \end{cases} = \min(9, 12) = 9$$

$D(2,4) = 8$

$$\text{Cost}(2,4) = \min \{ K = 8 : 11 + 7 \} = 18$$

$D(2,5) 8$

$$\text{Cost}(2,5) = \min \begin{cases} K = 7 : 11 + 5 \\ K = 8 : 8 + 7 \end{cases} \quad \min(16, 15) = 15$$



Topic : Dynamic Programming :(DP)

Stage-1

$$\text{Cost}(1,1) = \text{Min}\begin{cases} K = 2: \ C(1,2) + \text{Cost}(2,2) \\ K = 3: \ C(1,3) + \text{Cost}(2,3) \\ K = 4: \ C(1,4) + \text{Cost}(2,4) \\ K = 5: \ C(1,5) + \text{Cost}(2,5) \end{cases}$$

$$= \text{Min}\begin{cases} 2: \ 9 + 7 \\ 3: \ 7 + 9 \\ 4: \ 3 + 18 \\ 5: \ 2 + 15 \end{cases}$$



Topic : Dynamic Programming :(DP)

$$= \text{Min} \left\{ \begin{array}{ll} K & \downarrow \\ 2: & 16 \\ 3: & 16 \\ 4: & 21 \\ 5: & 17 \end{array} \right\} = 16$$

$D(1, 1) = \{2 \text{ or } 3\}$



Topic : Dynamic Programming :(DP)

#Q. What is the path that gives minimum cost?

$D(1,1) = 2$

$D(2,2) = 7$

$D(3,7) = 10$

$D(4,10) = t$



$(1 \rightarrow 2 \rightarrow 7 \rightarrow 10 \rightarrow t)$

OR path-2

D(1,1) = 3

D(2,3) = 6

D(3,6) = 10

D(4,10) = t

1 → 3 → 6 → 10 → t

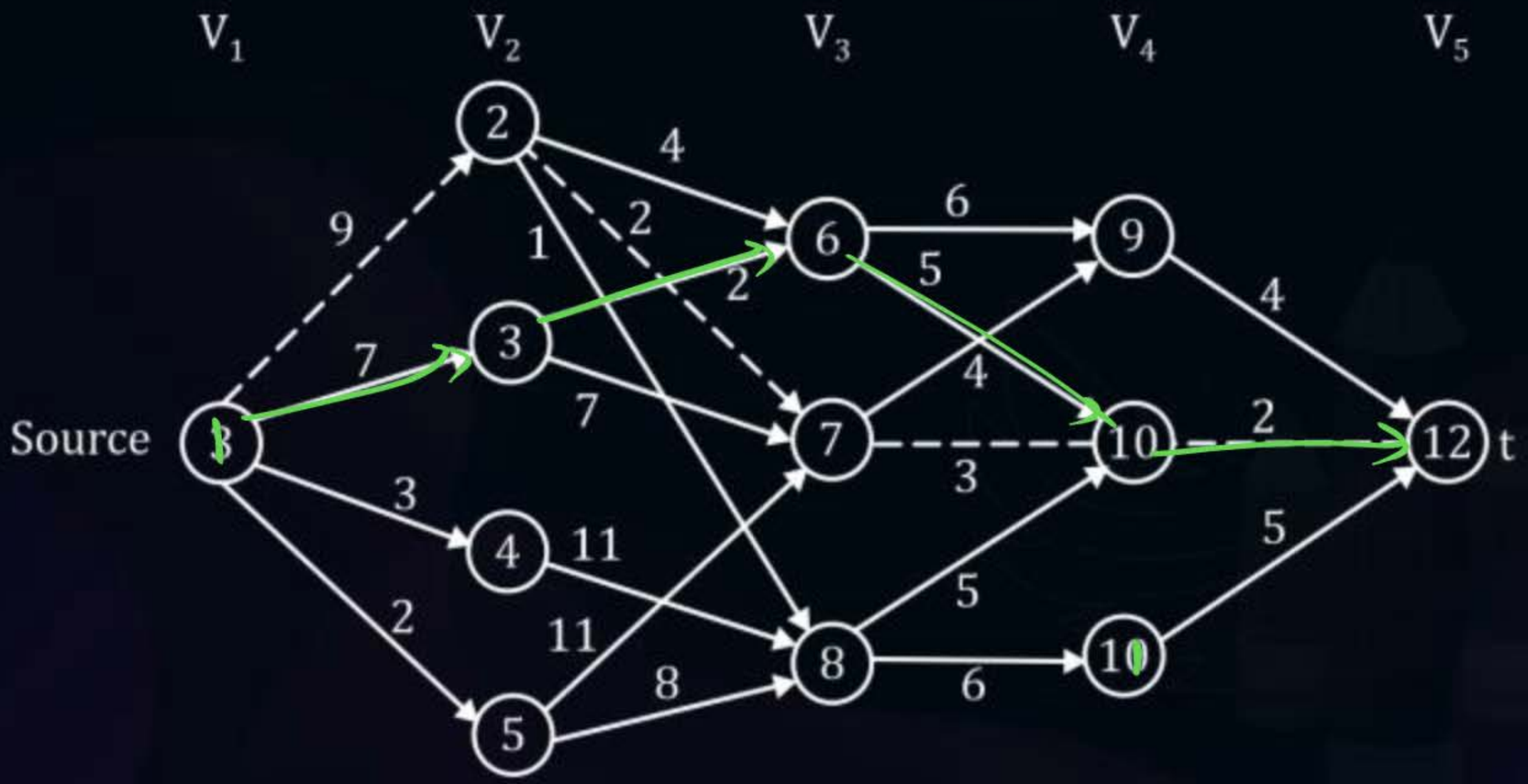
16





Topic : Dynamic Programming :(DP)

E.g. 2.





Topic : Dynamic Programming :(DP)

PYQ on DP:

1. Consider 2 strings:

   A = "qpqri"

   B = "pq pr qr p"

   Let x be the length of LCS between A and B, and let y be the number of such LCS between A and B.

14

then $x+10y = ?$

Soln:- A = qpqrri ——— 5

B = pqprqrp ——→ 7

5X

4

len(LCS) = 4 = x

Count of LCS = qpqr ⇒ 1 = y

x+10y

= 4+1×10 = 14

HW → Draw Recursion Tree

(Top Down)



Topic : Dynamic Programming :(DP)

Same question A: q pq rr → 5

B: pq prq r p → 7

Top Down

PYQ

len(LCS) = ?

5 ×

4 ✓

Ans - 4

→ (Top Down)

How many such LCS?

HW



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Multi - Stage Graph

↳ Top Down

↳ Bottom Up



THANK - YOU